

Experiment 'p_13_1_recursive_rectangle_area' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_13_1_recursive_rectangle_area

Natural language statement: Write a recursive method that takes the width and height of a rectangle and returns its area. Call the method with a width that is one less than the original.

Method signature: p_13_1_recursive_rectangle_area(width: nat, height: nat) returns (area: nat)

Ensures

- `area == width * height`

Requires

- `1 <= width`
- `1 <= height`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a recursive method that takes the width and height of a rectangle and returns its
→ area. Call the method with a width that is one less than the original.

The signature should be:

p_13_1_recursive_rectangle_area(width: nat, height: nat) returns (area: nat)

The method should respect the following contract:

requires 1 <= width, requires 1 <= height, ensures area == width * height

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnASeZp11atykKP2Kwg1ltNhME52V

Model: gpt-5.5-2026-04-23

Created at: 1780610656

Finish reason: stop

Usage: 322 tokens in, and 584 tokens out

```
{
  if width == 1 {
    area := height;
  } else {
    var smallerArea := p_13_1_recursive_rectangle_area(width - 1, height);
    area := smallerArea + height;
  }
}
```

Final Program

```
method p_13_1_recursive_rectangle_area(width: nat, height: nat) returns (area: nat)
  requires 1 <= width
  requires 1 <= height
  ensures area == width * height
{
  if width == 1 {
    area := height;
```

```
} else {  
  var smallerArea := p_13_1_recursive_rectangle_area(width - 1, height);  
  area := smallerArea + height;  
}  
}
```

Total Token Usage

Input tokens: 322

Output tokens: 584

Reasoning tokens: 512

Sum of 'total tokens': 906

Experiment Timings

Overall Experiment started at 1780610645684, ended at 1780610694679, lasting 48995ms (49.00 seconds)

Iteration #1 started at 1780610645781, ended at 1780610694673, lasting 48892ms (48.89 seconds)